

**AMENDMENT NO. 1 JANUARY 2008
TO
IS 12592 : 2002 PRECAST CONCRETE MANHOLE
COVER AND FRAME — SPECIFICATION**

(First Revision)

(Page 1, clause 3.1) — Substitute the following for the existing against grade Medium Duty:

Grade	Grade Designation	Type/Shape of Cover
Medium Duty	MD-10	Rectangular, Square, Circular

(Page 3, Table 1) — Substitute the table given on page 2 for the existing table.

(Page 4, clause 7.3.1) — Substitute the following for the existing:

‘The curing shall be done as per IS 456.’

(Page 4, clause 7.4.2, para 2, second sentence) — Substitute the following for the existing:

‘The ring made out of mild steel flat shall be given anticorrosive treatment by hot dip galvanizing.’

(Page 4, clause 8, para 1) — Substitute the following for the existing:

‘The minimum diameter of round steel rod used as lifting device shall be 10 mm for light duty, 12 mm for medium duty and 16 mm for heavy and extra heavy duty covers. The lifting device shall be made of naturally corrosion resistant metal rods or shall be protected from corrosion by hot dip galvanizing.’

(Page 4, clause 9.3) — Substitute the following for the existing:

‘The individual units shall be subjected to a test load as specified in Table 2 and in accordance with the method described in Annex C, satisfying the requirement specified therein. Also, the permanent set shall not exceed the requirement given in Annex C.’

(Page 7, Fig. 2) — Substitute the following for the existing figure:

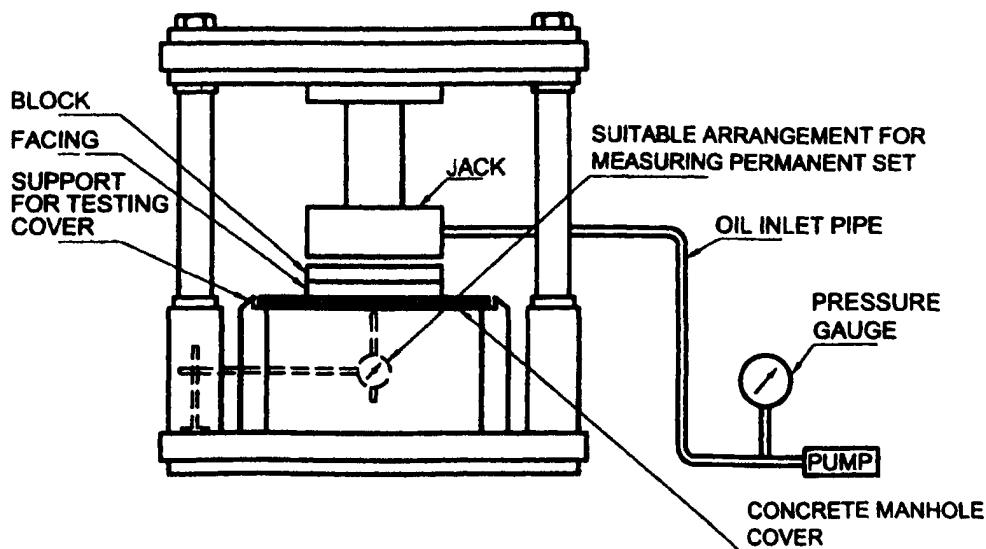
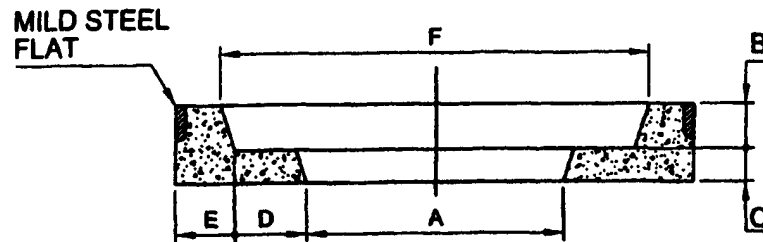


FIG. 2 ARRANGEMENT FOR LOAD TEST OF MANHOLE COVER

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Table 1 Dimensions of Frame
(Clause 5.2)
All dimensions in millimeters.



Grade Designation (1)	Description (2)	Clear Opening in Frame, A (3)	B (4)	C (5)	D (6)	E Min (7)	F Nominal (8)
LD-2.5	Light Duty Rectangular	450 × 600	50	50	50	50	559 × 709
LD-2.5	Light Duty Square	400 × 400 450 × 450	50 50	50 50	50 50	50 50	509 × 509 559 × 559
LD-2.5	Light Duty Circular	370 450 500 560	50 50 50 50	50 50 50 50	50 50 50 50	50 50 50 50	479 559 609 669
MD-10	Medium Duty Rectangular	450 × 600	70	50	50	50	560 × 710
MD-10	Medium Duty Square	450 × 450 600 × 600	70 70	50 50	50 50	50 50	560 × 560 710 × 710
MD-10	Medium Duty Circular	450 500 560 600	70 70 70 70	50 50 50 50	50 50 50 50	50 50 50 50	560 610 670 710
HD-20	Heavy Duty Rectangular (Scraper)	900 × 450	90	75	75	75	1 060 × 610
HD-20	Heavy Duty Square	560 × 560	90	75	75	75	720 × 720
HD-20	Heavy Duty Circular	450 500 560 600	90 90 90 90	75 75 75 75	75 75 75 75	75 75 75 75	610 660 720 760
HD-20	Heavy Duty Lamphole	350	90	75	75	75	510
EHD-35	Extra Heavy Duty Rectangular	900 × 560	100	75	75	75	1 060 × 720
EHD-35	Extra Heavy Duty Square	560 × 560	100	75	75	75	720 × 720
EHD-35	Extra Heavy Duty Circular	450 500 560 600	100 100 100 100	75 75 75 75	75 75 75 75	75 75 75 75	610 660 720 760

NOTES

1 Tolerance on C shall be ± 5 mm, tolerance on A, B, D and E shall be $\begin{matrix} +5 \\ -0 \end{matrix}$ mm.

2 For facility of removing the manhole cover suitable upward taper not more than 5° may be provided to the inner periphery of the frame.

3 If required for the removal of the moulds suitable taper not more than 5° can be given at the lower inner periphery of the frame (see figure).

4 Dimension of cover is covered in 5.2. The dimension F for frames is indicative only and has been arrived at based on the nominal values of dimensions A, B and D and a maximum taper of 5° on the inner periphery of the frame, wherever provided (see Note 2), subject to the provisions in 5.2.

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(Page 7, Annex C, clause C-1.2.1, para 2, first sentence) — Substitute the following for the existing:

‘The load shall be applied gradually at the rate of approximately 1 kN/s to 5 kN/s up to 2/3 of the test load.’

(Page 7, Annex C, clause C-1.2.1, para 3, second sentence) — Substitute the following for the existing:

‘The permanent set shall not exceed the following values:

For LD 2.5 and MD 10 — 1/100 time the diameter of the largest circle that can be inscribed in the clear area of the frame as shown in Fig. 3

For HD-20 and EHD-35 — 1/300 time the diameter of the largest circle that can be inscribed in the clear area of the frame as shown in Fig. 3

The permanent set shall be measured to an accuracy of 0.1 mm. Further, no crack wider than 0.2 mm shall appear in the concrete after this test at bottom side of cover.’

(Page 7, Annex C, clause C-1.2.2) — Substitute the following for the existing:

‘Immediately after the test according to C-1.2.1, the load shall be applied at the same rate as given in C-1.2.1, till the full test load is achieved. The test load shall be maintained for a period of 30 ± 2 s. There shall be no loss of adhesion between the concrete and the reinforcing steel at the end of the test.’